

fect. Either, they could use head-tracking to ensure that each of the viewer's two eyes sees a different image on the screen, or they could display multiple views so that the display does not need to know where the viewers' eyes are directed. Most 3D television prototypes and products available today are based on the use of lenticular lenses or parallax barriers.

The first technique involves the use of special LCD screens, the display surface of which is pasted with an array of tiny cylindrical plastic lenses called lenticules. These lenticules magnify the image shown to the viewer, and provide an effect similar to what you get when you view a cylindrical image—that is, your left and right eyes would see two different 2D images. The brain combines these to form one 3D image. Well, you do not have to wear glasses, but you might end up having to wear a seat belt, because lenticular lenses are dependent on where you sit, and even budging a little from the preferred spot would result in distorted images. A television would generally have more than one such spot, so multiple people can watch the show.

Companies like Sharp and LG are, however, pursuing the parallax barrier method, as it is more viewer-friendly and allows for both 3D and regular 2D viewing. Televisions that use this method would have a fine layer of liquid crystals placed in front of the screen. This layer, known as the parallax barrier, has slits in appropriate positions to enable the transmission of alternating images to each eye of the viewer when a slight voltage is applied to it. These images are slightly differ-

What happened to those old glasses?

The first 3D movie was made in 1922. For a long time after that, 3D viewing continued to be based on crude, coloured 3D glasses. These glasses were very inexpensive—you could buy one at the theatre for less than Rs 10, and it was nothing but a cardboard glass with two coloured cellophane papers for the two lenses. There were, of course, better models, but the idea was very simple.

The technique used by these old-fashioned coloured 3D glasses is called complementary colour anaglyphs. Basically, the lenses use filters to block certain colours from each eye. Typically, these use red and cyan lenses (or paper!) to pass red to one eye and blue and green to the other. When introduced, this method was capable of working only with B&W images, but latest methods allow for colour viewing as well, although the colour will be a bit dull. It is this colour loss that makes this technique inferior to others such as polarisation, and also causes nausea and headache in some viewers!

ent from each other and can be combined by the brain to form a 3D image. However, this method also requires the viewers to sit at certain spots.

The parallax barrier can be easily switched on or off using the television's controls, thereby enabling the TV to juggle between 3D and 2D viewing.

While such glasses-free 3D technology seems futuristic, some brands have already started launching, or at least promising, auto-stereoscopic models. Some are even more advanced. Toshiba's new launch, for instance, involves multi-parallax implementation. That is, instead of a barrier that telecasts just two alternating images to the viewer, this technology combines nine such images to create an extremely realistic 3D image.

"The glasses-free technology is under research currently as it requires the viewer to sit at a certain distance and certain angles to experience the 3D effect. Based on currently available technologies, we think it's optimal for us to provide TVs with glasses so that our customers can enjoy high picture quality. Meanwhile, Sony is also

working on the research and development of glasses-free 3D TVs," says Kimura.

3D-ready, but content starved

Most of the TVs in the market today are '3D-ready', and not exclusively 3D. That is, they are both 2D- and

3D-capable. So if you are prepared to pay a 30 to 40 per cent premium over normal 2D LCD televisions, you could buy a 3D television, and continue to use it as a regular TV also! For 3D experience, you have to buy the required glasses, a 3D-compatible disk player and some 3D movies, a 3D set-top box for subscribing to some 3D channels (if and when they come), or seek 3D content on the Web. A variety of 3D players, including DVD and Blu-ray ones, are available today.

That said, the real stumbling block for 3D, at the moment, seems to be the availability of content. There are only a few movies, and even less channels, available in 3D format.

"The popularisation of 3D technology depends largely on the availability of 3D content," says Kimura. The optimistic Sharma feels that the future is going to be quite bright: "Despite the glasses, the 3D viewing experience is very exciting as proved by the *Avatar* film. The *FIFA World Cup* coverage has shown the world the tremendous impact of live viewing of games in 3D. Most of the popular film-makers and big banners have 3D plans. Manpower is being trained for creation of content, and broadcasters have also expressed their interest." Some of the Indian Premier League (IPL) matches were also screened in 3D at a few multiplexes. Snippets were also streamed online.

Movies. *Avatar* has certainly improved the optimism in the industry with regards to 3D films, and there seem to be more than 15 3D



3D glasses